

Gray Spring

A 20 - 044	44	452	4.4	1989	5.3	2387	5.9	2685	6.6	2983	8.9	25
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- 6.6mm max compression (from datasheet)
- 17.24 deg ankle angle (from CAD)
- 452 N/mm (from datasheet)
- 62 Nm peak moment (code below)
- 206 Nm/rad peak stiffness

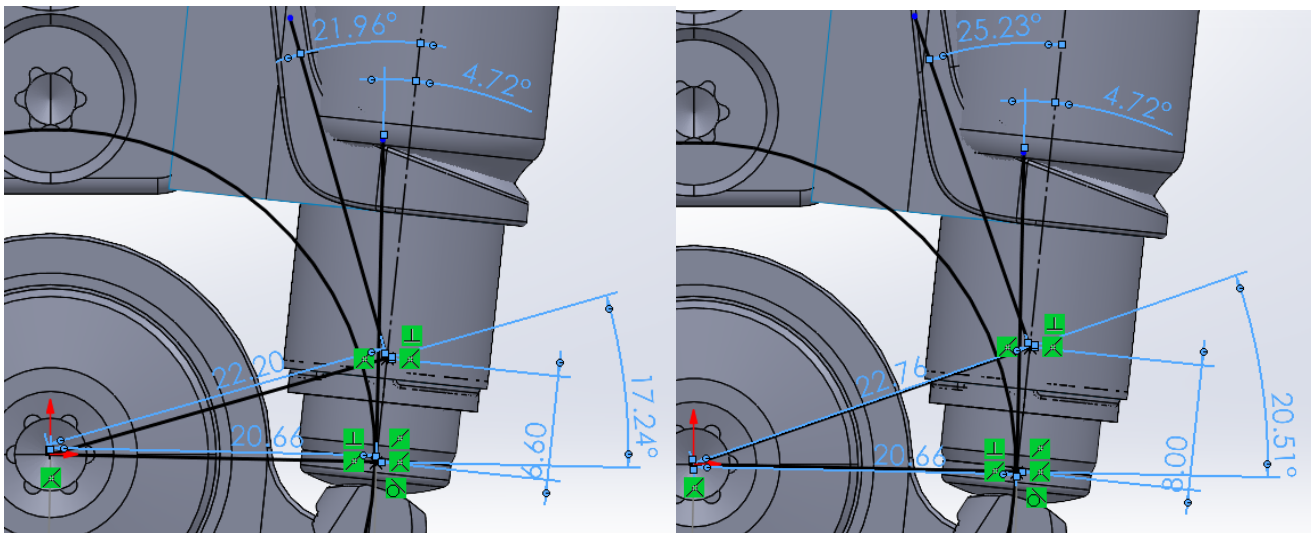
Yellow Spring

G 20 - 032	32	224	5.4	1210	6.4	1434	7.2	1613	8.0	1792	9.4	50
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- 8 mm max compression (from datasheet)
- 20.51 deg ankle angle (from CAD)
- 224 N/mm (from datasheet)
- 37 Nm peak moment (code below)
- 103 Nm/rad peak stiffness

VSO Linear Cam

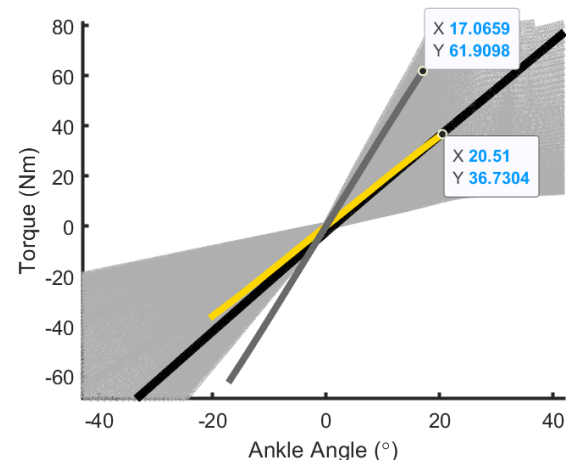
- Peak stiffness is 232% stiffness of yellow tango
- Peak stiffness is 116% stiffness of gray tango

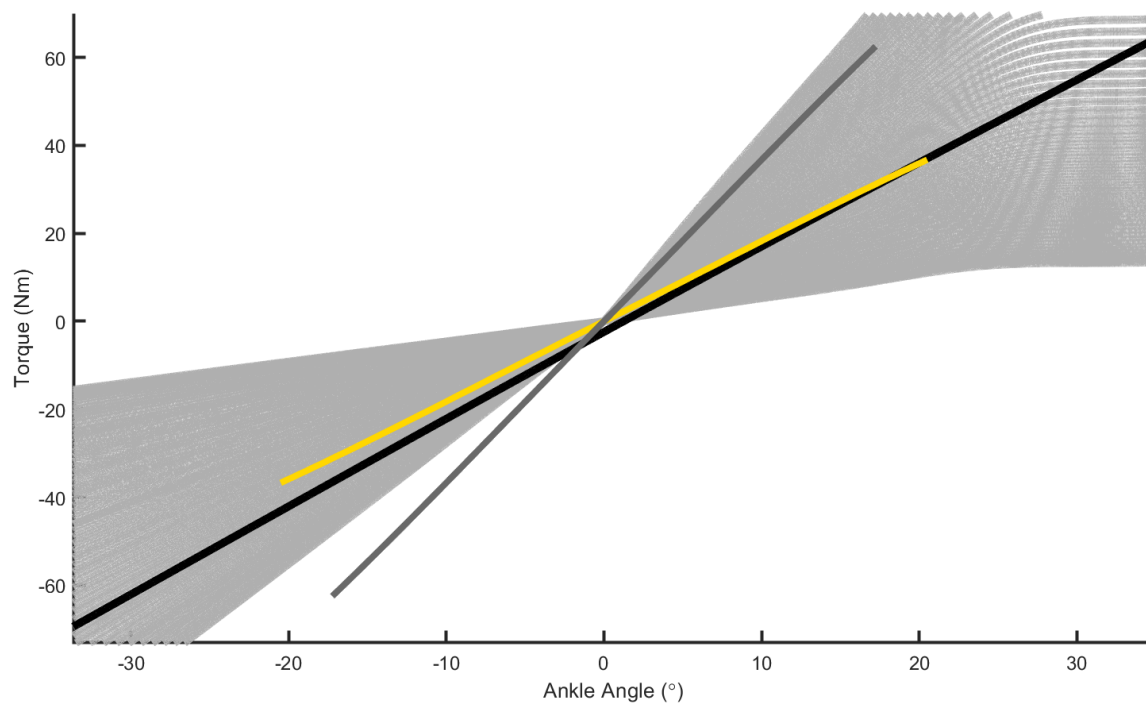


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thetaD_tango(1,:) = linspace(0,20.51,100);
thetaP_tango(1,:) = linspace(-20.51,0,100);
thetaD_tango(2,:) = linspace(0,17.24,100);
thetaP_tango(2,:) = linspace(-17.24,0,100);
lever_arm_tango(1,:) = linspace(0.02066,0.02276,100);
lever_arm_tango(2,:) = linspace(0.02066,0.0222,100);
x_tango(1,:) = linspace(0,8,100)*mm2m; %yellow
x_tango(2,:) = linspace(0,6.6,100)*mm2m; %gray
k_tango = [223,460]./mm2m; %N/m
phi_tango(1,:) = linspace(4.72,25.23,100);
phi_tango(2,:) = linspace(4.72,21.96,100);
for t=1:length(k_tango)
    if(t==1)
        color = [255,215,0]./255;
    else
        color = [105,105,105]./255;
    end
    force_spring_tango = k_tango(t).*x_tango(t,:);
    moment_tango = force_spring_tango.*cosd(phi_tango(t,:)).*lever_arm_tango(t,:);

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Black: estimated torque-angle function for plastic cam tested at Ottobock

Yellow: yellow spring tango

Dark Gray: gray spring tango

Light gray lines: VSO at different spring support locations with linear cam